CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

FORWARD CHAINING

Aim:

To implement the Forward Chaining algorithm in Prolog.

Objective:

To derive new facts from an initial set of facts by repeatedly applying rules.

Algorithm:

1. Start.
2. Define initial facts.
3. Define rules.
4. Check whether all conditions of a rule are satisfied.
5. If satisfied, derive the conclusion.
6. Add the new fact to the knowledge base.
7. Repeat until no new facts can be derived.
8. Stop.

Flowchart:

START

| v

Load Initial Facts

| v

Check Rules

| v

Conditions Satisfied?

/ \

Yes No

| |

Derive Fact Check Next Rule

| v

Add Fact to KB

| v

New Fact Generated?

/ \

Yes No

| |

Repeat STOP Prolog Program:

```
% Forward Chaining
fact(fever). fact(cough).
rule([fever, cough], flu). rule([flu], doctor_visit). rule([doctor_visit], treatment).
forward :-collect_facts(Facts),
forward_chain(Facts, FinalFacts), write('Final Facts: '), write(FinalFacts),
nl.
collect_facts(Facts) :-findall(Fact, fact(Fact), Facts).
forward_chain(Facts, FinalFacts) :-findall(
Conclusion,
(
rule(Conditions, Conclusion), all_present(Conditions, Facts),
\+ member(Conclusion, Facts)
),
NewFacts
), (
NewFacts = [] -> FinalFacts = Facts
;
append(Facts, NewFacts, UpdatedFacts), forward_chain(UpdatedFacts, FinalFacts)
).
all_present([], _).
all_present([H|T], Facts) :-member(H, Facts), all_present(T, Facts).
```

Query

?- forward.

Output

Final Facts: [fever,cough,flu,doctor_visit,treatment]

true.

Result:

Thus, Forward Chaining is successfully implemented to derive new facts from the knowledge base.